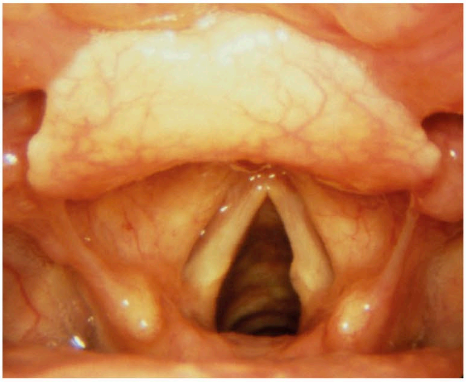
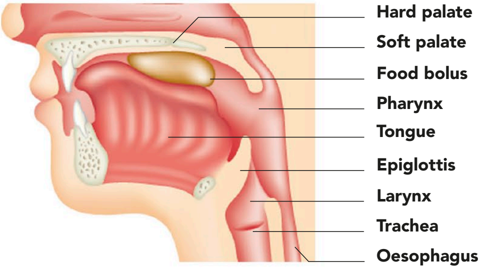
SWALLOWING

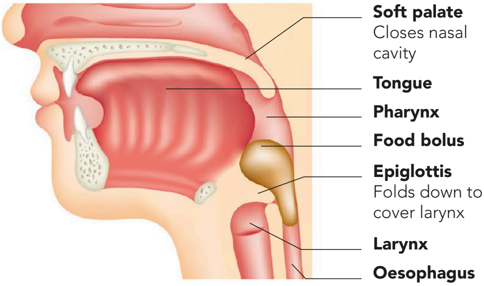
The process of swallowing begins as a voluntary action, when the rear of the tongue pushes a bolus of food to the back of the mouth. To swallow a solid item such as a tablet without chewing demands concentration. It is easier to swallow a tablet with water, as drinks are usually gulped straight down after entering the mouth. Automatic reflexes control subsequent stages of swallowing, as the muscles of the throat contract and move the bolus rearward and down, and squeeze it into the top of the oesophagus. A flap of cartilage known as the epiglottis prevents food going down "the wrong way" into the larynx and the trachea, where it would cause choking.



VIEW INTO THE LARYNX
The pale, leaf-like flap of the epiglottis is visible at the top of this image. Immediately below it is the inverted "V" of the vocal cords.



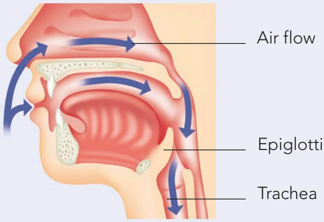
1 PHARYNGEAL STAGE
Before the food bolus reaches the back of the mouth, the epiglottis is raised in its normal position.



2 OESOPHAGEAL STAGE
The larynx rises to meet the tilted epiglottis, closing the trachea. The soft palate lifts to close the nasal cavity. The bolus is pushed down the oesophagus.

BREATHE OR SWALLOW

The pharynx is a dual-purpose passageway: for air when breathing, and food, drink, and saliva when swallowing. Nerve signals from the brain operate the muscles of the mouth, tongue, pharynx, larynx, and upper oesophagus to prevent food from entering the trachea. If food is inhaled, irritation of the airway triggers the coughing reflex, which expels the inhaled particles and prevents choking. The complex muscle movements involved in swallowing are a voluntary reflex, and they also occur when solid matter contacts touch sensors at the back of the mouth.



DUAL INTAKE
Breathing occurs through the nose or the mouth. The passageways of both meet at the throat, and air flows into the trachea.